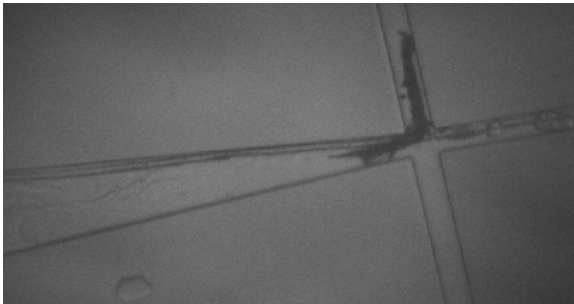
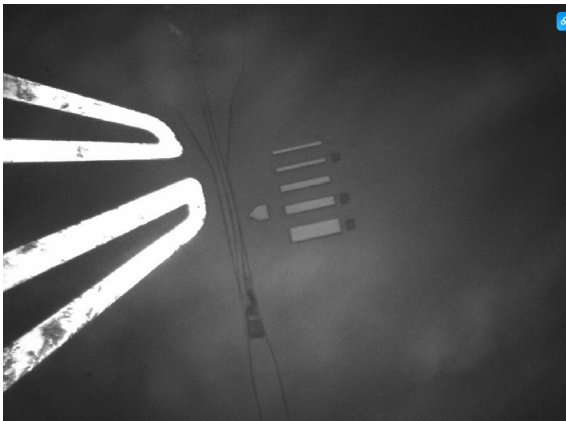


Lab Notebook (Microfluidic Platform)

July

Optimization of Droplet-Generation Chip Clogging

Frequent clogging increased chip consumption; the clogging causes of the generation chip were analyzed and improvements were made.



Clogging material type	Possible appearance	Main hazards	Common sources
Small particulate impurities	Black dots, bright spots, small block-like	Aggregate to block narrow channels, change the flow rate, and destabilize droplet size	Unfiltered reagents, airborne dust, PDMS punching debris, glass debris

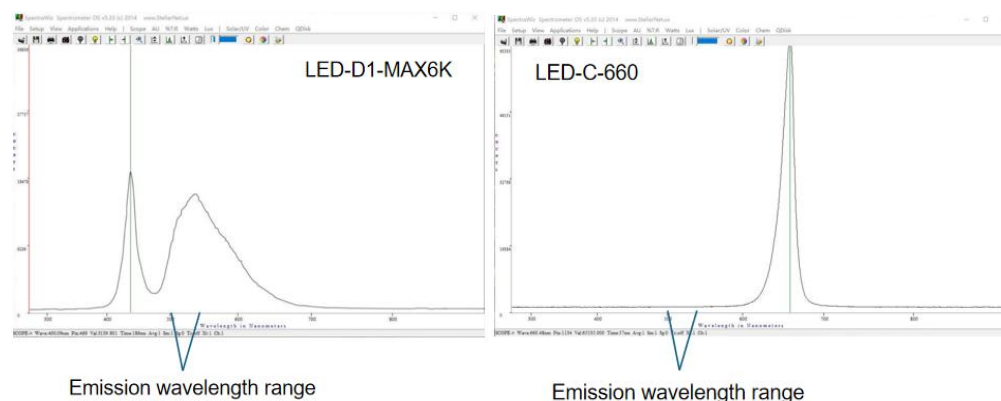
	deposits		
Fibrous impurities	Thin, elongated, thread- or hair-like shapes spanning the channel	Most likely to form “bridge-type clogging”; subsequent particles keep hanging on them, making clogging progressively worse	Wipes, lint-free cloths, cotton swabs, clothing fibers, tubing-cutting residue, detached filter membranes
Colloids / agglomerates	Lumpy, translucent, possibly angular	Directly block small channels, affecting fluorescence detection and DEP sorting	Bacterial aggregation, protein/polysaccharide precipitation, medium residue, surfactant destabilization, PDMS debris
Cell / bacterial clumps	Deposited in clumps, often at narrow junctions and bifurcations	Cause sudden increases in flow resistance, uneven droplet encapsulation, and abnormal fluorescence signals	Excessively high bacterial concentration, viscous debris from dead cells

The main clogging cause: the distilled water used for the aqueous phase was not sufficiently pure. Improvements: 0.22 μ m membrane filtration and widening the flow channel.

Optical modification: switched to a 700 nm LED for top illumination of the chip, with an ND filter in front of the camera and a 535 nm filter in front of the PMT.

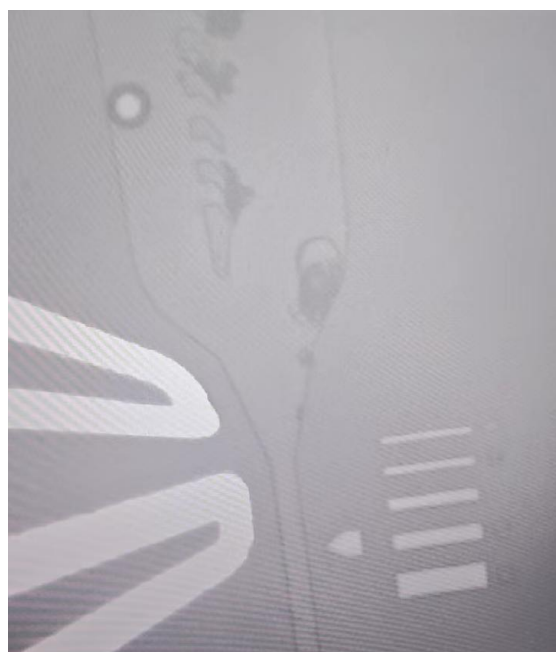
Optical-Path Structure Optimization

To solve the overexposure problem of the microscopic observation camera, three optimizations to the optical path were planned: replace the LED with top illumination of the chip, add an ND attenuation filter in front of the camera to reduce light intensity, and add a 535 nm filter in front of the PMT to select the target fluorescence.



Initial Test of the Sorting Function

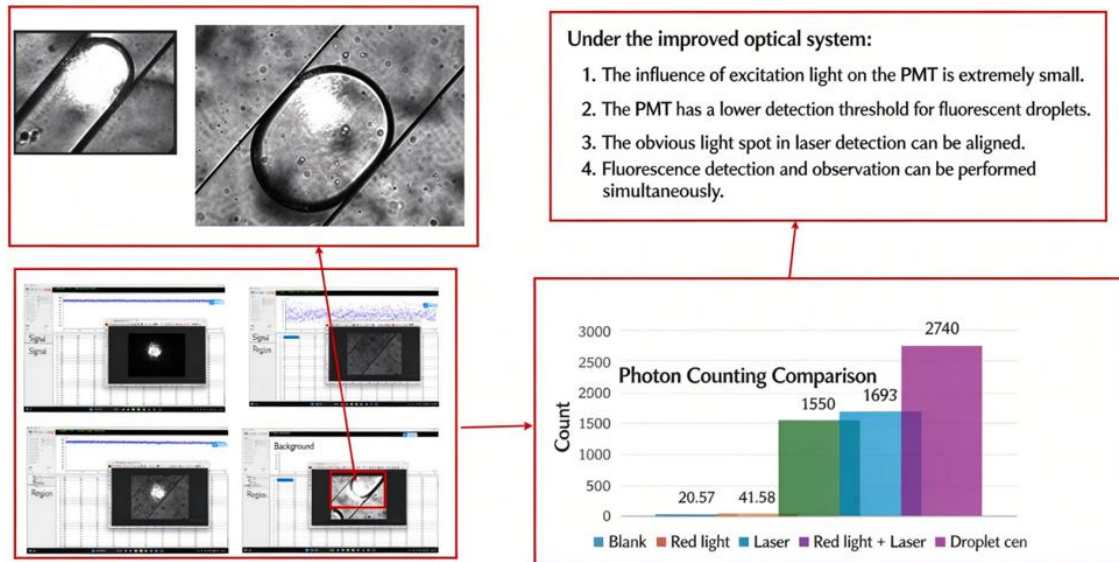
By continuously turning on the sorting switch, it was confirmed that sorting operations could be performed; the high-voltage amplifier model was also confirmed, and all hardware was finalized.



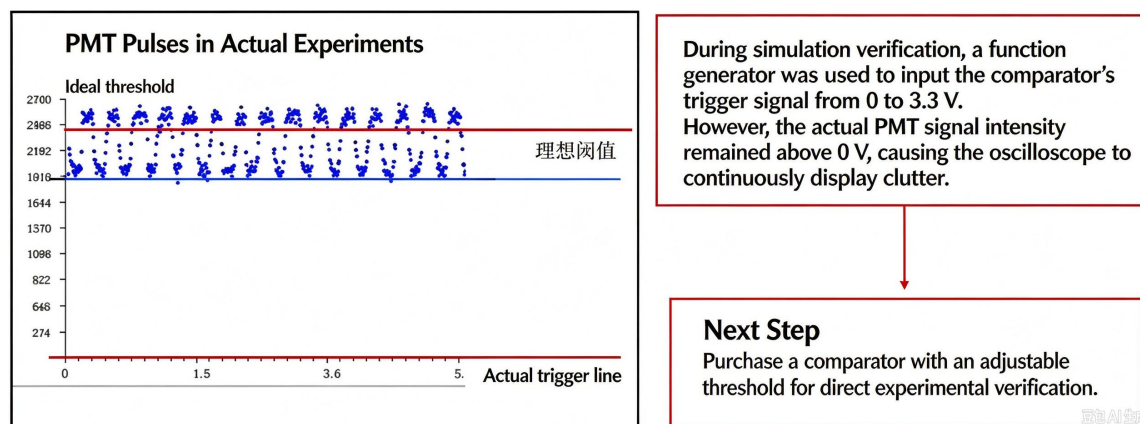
Video 8 — Sorting-switch test

Resolving the Overexposure Problem at the Observation End

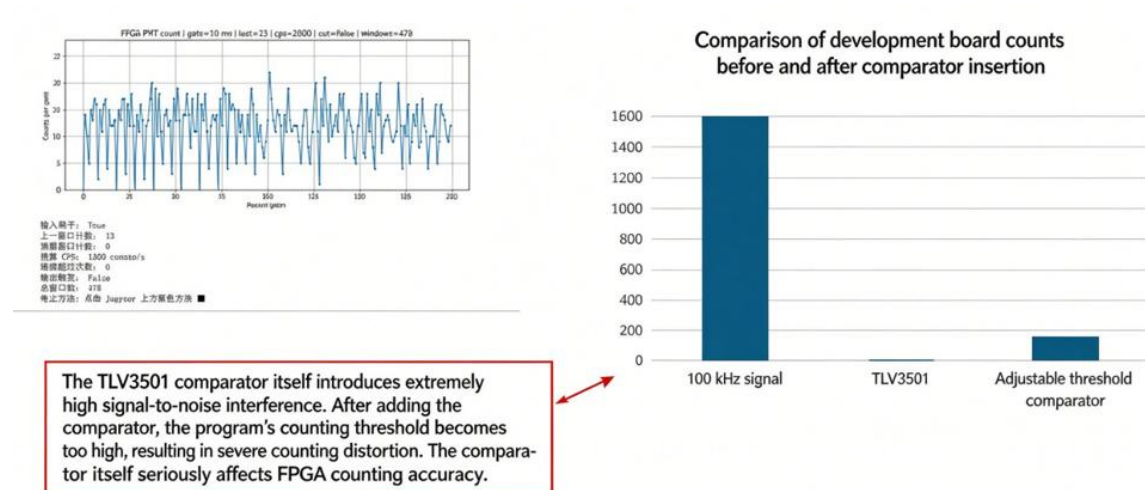
Switched to a 700 nm LED for top illumination of the chip, added an ND filter in front of the camera to attenuate light, and a 535 nm filter in front of the PMT.



Testing and Optimization of the Photon-Detection Counting Capability



Because the PMT signal baseline was always above 0 V, the comparator kept triggering and produced spurious signals. An adjustable-threshold comparator was then used for experimental validation, which revealed that the comparator interfered with signal acquisition.



However, connecting the PMT directly to the development board could not capture the 10 ns ultra-narrow pulses, limited by the minimum pulse width that the PYNQ-Z2 can capture. Two solutions were therefore proposed: (1) use an SN74LVC1G123 chip to stretch the pulse width; and (2) change the hardware chain to PMT-CH297-FPGA, using the CH297 counter head to acquire pulses.

All videos referenced in this document are available for download at the bottom of our team' s Notebook page.